Basel Ahmed Sardah

Computer Science graduate (GPA 4.79/5.00) engineering distributed systems, low-latency edge AI inference pipelines, and production mobile applications. Experienced in architecting event-driven FastAPI microservices, optimizing on-device database architectures, and maintaining commercial Flutter applications published on the Apple App Store.

BaselSardah@outlook.com +966570031217 https://www.linkedin.com/in/BaselCS/

WORK EXPERIENCE

Beaee Platform

Software Engineer Intern Jan/2026 - Jul/2026

- Maintained the commercial Flutter mobile application published on the Apple App Store, resolving runtime defects and optimizing UI rendering performance across release cycles.
- Automated stakeholder communication and HR scheduling workflows by engineering a Python script that ingests CSV datasets for scheduled WhatsApp dispatches, saving ~15 operational hours daily.
- Standardized client-side state architecture using the Stacked framework (MVVM) and GetIt dependency injection, cleanly isolating business logic to prevent UI mutation collisions.
- Restructured relational database schemas and data normalization layers, accelerating complex analytical SQL query performance while expanding schema scalability.
- Containerized backend API services and database environments using Docker, establishing strict development-to-production parity to eliminate deployment discrepancies.
- Directed sprint task allocation across the SDLC as Technical Team Lead, translating business requirements into granular technical deliverables in coordination with executive management.

PROJECT

Al Mihbara: Real-Time Multi-Client Interactive Platform | FastAPI, Redis, PostgreSQL, Pixi.js, XState 2026

- Architected an event-driven microservice backend utilizing FastAPI, Redis Pub/Sub, and Server-Sent Events (SSE) to deliver low-latency real-time state synchronization across concurrent web clients.
- Engineered deterministic client-side game state transitions using XState v5 finite state machines, eliminating race conditions during rapid multi-user state mutations.
- Enforced strict multi-tenant data access security and database integrity by configuring PostgreSQL and Supabase Row Level Security (RLS) policies.

QASSAS: Industrial Visual Anomaly Detection Pipeline | Python, PyTorch, Diffusion Models, OpenCV 2026

- Engineered an unsupervised visual anomaly inspection pipeline integrating fine-tuned Stable Diffusion v1.5 with Orthogonal Fine-Tuning (OFT) LoRA adapters and a 200-tree Isolation Forest classifier, achieving a 97.08% defect recall rate.
- Constructed a GPU-accelerated 5D metric extraction factory (L1, L2, MS-SSIM, VGG-based LPIPS, 16x16 Max-Patch) using PyTorch TensorFloat-32 and FP16 Automatic Mixed Precision to execute sub-second per-sample vectorization.
- Bypassed GPU Out-Of-Memory (OOM) failures on 1024x1024 inputs by formulating a 4-quadrant patch decomposition and spatial max-pooling pipeline, reducing peak VRAM footprint by 75%.

FruitBot: Real-Time Low-Latency Automation System | Python, TensorRT, Linux, OpenCV 2026

- Architected an asynchronous multi-threaded processing pipeline sustaining 60+ FPS at sub-10ms latency by decoupling frame acquisition, model inference, and video persistence into background daemon threads.
- Eliminated shell subprocess overhead by dispatching cursor actions directly through persistent Unix Domain Sockets to the Wayland compositor.
- Optimized memory management to O(1) allocation cost by implementing zero-copy buffer casting directly into pre-allocated contiguous NumPy arrays, preventing heap re-allocation thrashing.

Salad of Achievement: Productivity Application | Flutter, ObjectBox NoSQL, Provider 2025

- Engineered an on-device NoSQL persistence engine using ObjectBox C-bindings across 5 core relational models, achieving sub-millisecond query execution by replacing disk-bound database calls with direct C-pointer memory operations.
- Eliminated background timer drift and OS thread throttling by implementing an absolute epoch-timestamp delta tracking algorithm (TimerLogic) paired with Flutter WidgetsBindingObserver lifecycle hooks.
- Maintained a continuous 60 FPS UI rendering target across 7 interactive screens within a monolithic state/MVC-hybrid pattern by isolating updates with Provider and offloading database write transactions through microtask execution queues.

EDUCATION

- Bachelor of Computer Science — King Faisal University | GPA: 4.79 / 5.00 (Top Standing) | 2021 – 2026
- Certifications: Google Data Analytics Professional Certificate (2024) | Machine Learning Specialization & Advanced Learning Algorithms (Stanford / DeepLearning.AI, 2025) | Intro to Data Science in Python (Univ. of Michigan, 2025)

SKILLS

- Languages & Runtimes: Python, Dart, SQL, Bash, HTML5
- Backend & Architecture: FastAPI, REST APIs, Server-Sent Events, Redis Pub/Sub, Docker, MVVM, MVC, XState v5
- Databases & Storage: PostgreSQL, Supabase RLS, MySQL, SQLite, Firebase, ObjectBox NoSQL
- Frontend & Mobile: Flutter (iOS, Android, Web), Pixi.js, Responsive UI Design, State Management (Provider, Stacked, BLoC)
- Tools & Environments: Linux (Arch, Wayland, Unix Sockets), Git, GitHub, Uvicorn, uv, Agile/Scrum, Linear, Trello